Karan Upadhyay

Software Engineer • Full-Stack & Applied AI • React Native • Node.js • Python • LLM Agents

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secretJod | LinkedIn | LeetCode | Portfolio

EXPERIENCE

Jain Irrigation Systems Ltd (JISL) Jun 2026 – Present

Full-Stack Developer | Applied AI *Jalgaon, Maharashtra*

Anubhuti Parent Connect – iOS, Android & Web (sole developer) | *React Native, Expo, Node.js, MySQL, AWS*

- Took a school app from architecture to launch on the App Store, Google Play and web in 8 weeks, using Claude Code agents with MCP connectors to speed up building, testing and releases; used by about 200 parents and 15 teachers.
- Built a "My Child" album: a face-recognition model fine-tuned on consented school photos finds each child, even in group shots, at 96.4% precision and 91.8% recall, and sends unsure matches to a teacher.
- Made the slowest paths over 100x faster: album tags from 2,000 ms to 16 ms, file lookups from 1,222 rows read to 3, and an 82% smaller web bundle for faster first load on phones.
- Fixed real production issues, including a leak of classmates' personal details, an OTP sign-in bug, and duplicate records from double taps.
- Cleared Apple review and set up safe releases to AWS EC2 with backups, repeatable migrations, nginx and PM2.

AI-Driven Engineering | *Claude Code, tmux, MCP, multi-agent orchestration*

- Built a multi-agent workflow that runs Claude Code sessions in tmux as a team: a lead agent plans and splits the work, worker agents build in parallel, and I approve every production deploy.
- Built and deployed an MCP server serving 81 design themes to coding agents, and linked agents to Bitbucket and Gmail via MCP connectors.
- Used this setup to take 10+ internal tools from idea to working software in 4 months, while I owned the architecture, reviews and releases.

Internal Platforms – manufacturing, procurement & finance | *Next.js, PHP, MySQL, Prisma, n8n*

- Offline-first production app for the moulding plant: 196 machines, works without internet, syncs to MySQL and exports to SAP.
- Vendor contract platform and AI invoice review (Gmail, n8n, Gemini, SAP) with safeguards that prevent double payments and double posting.
- Used the Gemini and Claude APIs to turn invoice data into spend trends and anomaly alerts, with every number checked against source records.

SmallFare Aug 2025 – Oct 2025

Software Development Intern *Remote*

- Incorporated a 3D model viewer using React Three Fiber, replacing 100+ images with a single Blender model — cutting page size 450MB→7MB and backend load by 98%.
- Optimized e-commerce search by integrating Elasticsearch + Redis with AI-driven ranking, boosting query relevance and response speed.
- Implemented JWT auth (access + refresh tokens) and migrated 25+ endpoints to Prisma ORM, improving type-safety and reducing injection vulnerabilities by 30%.
- Remediated a critical security vulnerability preventing unauthorized admin access; optimized database query performance platform-wide.

PROJECTS

AQ Space – AI Learning Workspace | *Next.js, FastAPI, PostgreSQL + pgvector, Celery, RabbitMQ, Kafka, Redis* Code

- An AI study app: upload lectures, get timestamped notes with sources, and see which topics your playlist is missing and what to learn next.
- FastAPI backend with Celery workers, RabbitMQ for priority jobs, Kafka for events, pgvector for semantic search, and Google sign-in.
- Failed jobs resume where they stopped, so no AI call is paid for twice; caching cut dashboard load time from 410 ms to 38 ms.
- Measured AI quality: 87% correct concepts, 3.2% unsupported statements, and $0.42 per 30-minute video.

URL Shortener | *Go, React, PostgreSQL, Redis, Docker, GitHub Actions* Code

- Engineered a Go/Fiber and React URL shortener with API-key auth, per-link authorization, rate limiting and fail-closed SMTP email verification.
- Moved click analytics off the redirect path with Redis Streams and a consumer group; collision-free codes via Redis counter and SETNX.
- Built GitHub Actions CI/CD that publishes multi-architecture images to GHCR and runs CI-gated, health-checked deploys to a self-hosted runner behind Cloudflare Tunnel, with no manual steps after a push.
- Load-tested the live deployment with k6: 0% errors under sustained traffic; the rate limiter rejected 11 of 30 burst requests with Retry-After.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Go, SQL, Java, C/C++, PHP

Development: React Native, Expo, React, Next.js, Node.js, Express, FastAPI, MySQL, PostgreSQL, Redis, RabbitMQ, Kafka

AI Engineering: Claude Code, multi-agent orchestration, MCP servers and connectors, RAG, pgvector, LLM evaluation, fine-tuning

Cloud & DevOps: AWS EC2, nginx, PM2, Docker, GitHub Actions, Cloudflare Tunnel, Prometheus, Grafana

EDUCATION

LNCT, Bhopal – Master of Computer Applications (AI & ML), CGPA 8.5 2023 – Present

NMU University, Maharashtra – B.Sc. in Computer Science, CGPA 9.11 2019 – 2022

ACHIEVEMENTS

- 1st Prize for an AI & ML research paper at an international conference; 900+ LeetCode problems solved, top 35% globally.
- Flipkart GRID 7.0 Semifinalist (top 1% of 160,000+); Tata Imagination Challenge Semifinalist in 2023 and 2024 (312,000+ participants).
- Smart India Hackathon internal-round qualifier and team lead; Runner-up at GDG Bhopal; Top 5 at Web3Spell; Top 10 at Sheryians Hackathon.